Kuri Upload Specification

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The file the GIS team hands the developer, what the Admin Portal does with it, and how background cadastral auto-enrichment operates.

What changes

The upload workflow receives 12 essential GIS signals. All downstream QA and operational fields that previously cluttered spreadsheets have been moved into their dedicated portal stages.

> **Auto-Enrichment via TNGIS WFS & Revenue Master**
>
> In addition to the 12 uploaded columns, Kuri automatically enriches each record on upload by querying the **Tamil Nadu GIS (TNGIS) GeoServer WFS** (`cadastral_analysis:fmb_ulpin`). Through point-in-polygon ray-casting, Kuri derives the **Survey Number**, **Sub-division Number**, and **ULPIN**, while canonicalizing Revenue Village and Taluk names against the authoritative Tamil Nadu Revenue Master.

The most important signal retained is `area_sqm`. The eligibility rule is a total plot area over 1,500 sq ft (139.35 sq m). When imported, Kuri calculates `area_sqft` and pre-evaluates eligibility. Sub-threshold plots import with a `Not Eligible` pre-check flag (marked `warn`, not rejected outright), enabling analysts to quickly review corner plots or combined parcels without losing data.

Column specification

Twelve columns are supplied by the GIS team. The GIS team does not send a Site ID; Kuri mints a standard 6-digit Site ID (`CBE-XXXXXX`) on upload. A row failing a `reject` rule halts the entire batch. A row failing a `warn` rule imports flagged for QC review.

Column	Type	Required	Rule	On Failure
Site ID	text	not sent	Created by Kuri on upload as standard 6-digit `CBE-XXXXXX`. Never supplied in upload file.	reject
lat	decimal	yes	10.6 to 11.4 (Coimbatore district bounding envelope; prevents coordinate slippage).	reject
lon	decimal	yes	76.6 to 77.4 (Coimbatore district bounding envelope).	reject
Village	text	yes	Non-empty. Canonicalized against Tamil Nadu Revenue Master.	reject
Taluk	text	yes	Validated against Coimbatore taluk list. Resolves historical misattributions.	reject
District	text	optional	Defaults to `Coimbatore` if omitted. Validated against revenue hierarchy.	default
detection_type	enum	yes	One of `greenfield`, `infill`, or `long_term`.	warn
construction_stage	text	no	Stored as `detected_stage`, read-only. Never overwritten by field or QC	warn

Column	Type	Required	Rule	On Failure
			observations.	
area_sqm	decimal	yes	Must be > 0. Converted to sq ft for 1,500 sq ft threshold validation.	**reject**
model_confidence_score	decimal	yes	0.0 to 1.0. Seeds the QC queue ordering.	**warn**
priority_tier	enum	yes	`tier1`, `tier2`, or `tier3`. Governs inspection priority queue.	**warn**
meets_strict_construction_definition	boolean	yes	`TRUE` or `FALSE`.	**warn**
CONTRADICTS_planetscope_still_vegetated	boolean	yes	`TRUE` or `FALSE`. Flag indicating potential vegetation false positive.	**warn**
dtcp_building_type	text	no	Pre-fills *Type of Construction* where present (e.g. Residential, Commercial).	**warn**

What the portal works out for itself

Attributes derived automatically and consistently by Kuri on upload, eliminating manual calculation errors:

Derived Attribute	Derived From	Operational Benefit
Site ID (Unique ID)	Kuri Sequence Engine	Mints clean 6-digit `CBE-XXXXXX` format. Never alters after creation.
Cadastral Parcel (Survey & Sub-div)	TNGIS WFS GeoServer	Auto-queries `cadastral_analysis:fmb_ulpin` via point-in-ring geometry. Resolves Survey No, Sub-division, and ULPIN.
Google Maps Link	`lat` , `lon`	One fewer column to upload; guaranteed to never disagree with coordinates.
area_sqft	`area_sqm × 10.7639`	Single standardized conversion applied uniformly across all records.
Eligibility Pre-check	`area_sqft >= 1500`	Analyst opens record immediately knowing if plot meets client threshold.
Duplicate Flag	Live sites within 40m	Identifies existing sites before verifiers travel. (50m for multi-unit projects).
Formatted Address	Google Geocoding + Master	Structured street name, locality, taluk, and postal code from coordinates.

Import & Deduplication Rules

- **Site ID Standard & Immutable Guarantee:** The Site ID is minted once by Kuri as standard 6-digit `CBE-XXXXXX` (e.g., `CBE-518294`). An ID is never regenerated, reused, or reformatted. Legacy coordinate IDs are permanently maintained in `old_unique_id` .
- **Coordinate Collision Handling:** In rare cases where multiple sites occupy the same ~11m GPS cell, the secondary record receives a deterministic row suffix (e.g. `CBE-109482-769895-r298`) in `old_unique_id` , guaranteeing 100% 1-to-1 uniqueness.
- **Matching on Re-upload:** Rows matching an existing site within 40m update metadata without creating duplicate records. Changed details are logged, while identical rows are safely ignored.
- **Batch Integrity:** All-or-nothing execution. If any row violates a `reject` rule, the entire upload is halted with clear row-level error reporting.
- **Initial Status:** Status on arrival is strictly `QC Pending` . Field verifiers are never assigned at import; assignment requires QA approval.

What moves into the portal

These columns represent analytical and ground verification work, preserved with strict user audit attribution:

Column in Legacy File	Where It Belongs in Modernized Architecture
QC'er Name	Extracted automatically from the signed-in session; never typed.
Street View Details	Recorded in QC modal with coverage status and visual landmarks.
Google Earth / VHR Imagery	Recorded with satellite source (e.g. Cartosat, Maxar) and acquisition date.
Additional Details	Stored as structured QC comments.
Contact No	Stored with stakeholder role attribution (Owner, Contractor) and source URL.

Column in Legacy File	Where It Belongs in Modernized Architecture
Correct Coordinates	**Nearby Case:** Spawns a new child record with its own 6-digit Site ID linked through `Parent Site ID`, preserving the detection discrepancy.

Agreed Value Lists

Column	Allowed Values	Rule / Behavior
`detection_type`	`greenfield`, `infill`, `long_term`	Categorizes detection environment.
`priority_tier`	`tier1`, `tier2`, `tier3`	Orders the QC queue and assignment dispatch.
`audit_decision`	`CONFIRMED`, `NEARBY`, `NOT_OBSERVED`, `INCONCLUSIVE`	Standardized desk assessment outcomes.